

Project ID :

R25-040

1. Topic (12 words max)

AI based candidates recommending system for a company vacancy.

2. Research group the project belongs to

CEAI - Centre of Excellence in AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Recruitment is essential for organizational success, but traditional processes are often inefficient and biased. Manual CV screening and conventional interviews primarily focus on hard skills and listed qualifications, overlooking soft skills and behavioral attributes. These inefficiencies, highlighted by LinkedIn Talent Solutions (2023), hinder organizations from identifying the most suitable candidates for dynamic roles.

A significant challenge in recruitment is the sheer volume of applications. Manual screening becomes labor-intensive and inconsistent, with recruiters often relying on subjective judgments. Furthermore, technical assessments, while valuable, frequently lack the contextual realism to evaluate how candidates' skills translate to workplace scenarios (Landers & Marin, 2021). Soft skills, including communication, leadership, and emotional intelligence, are also challenging to measure objectively, as traditional interviews often suffer from unconscious biases (Chamorro-Premuzic & Frankiewicz, 2019).

Artificial intelligence (AI) offers transformative potential to address these challenges. Advanced tools, such as Natural Language Processing (NLP), enable automated CV parsing and analysis, extracting relevant details like education, skills, and job experience. AI models can rank candidates in real-time based on job requirements, reducing redundancy and ensuring fairness (IJCRT, 2023; IEEE, 2024)

Beyond CV screening, AI-driven simulations assess technical proficiency by replicating real-world tasks, providing insights into candidates' adaptability and problem-solving capabilities (Landers & Marin, 2021).

Behavioral analytics is another area where AI excels. Sentiment analysis and video analytics tools evaluate candidates' responses, capturing nuanced traits such as emotional intelligence and decision-making (Chamorro-Premuzic & Frankiewicz, 2019). Similarly, AI technologies assessing social traits analyze non-verbal cues like facial expressions and vocal tone, offering a holistic view of a candidate's interpersonal skills (Greenfield & Griffin, 2018)

The proposed research aims to develop an AI-based candidate recommendation system that integrates these advanced techniques. By integrating Natural Language Processing (NLP) for CV evaluation, machine learning for assessing technical competencies, and AI-driven models for behavioral and social trait analysis, the system performs the process of selecting the best candidate. This holistic approach enables organizations to identify the most suitable candidates efficiently, improving both hiring outcomes and organizational success.

References

- ❖ LinkedIn Talent Solutions. (2023). *Global Recruiting Trends Report*.
- ❖ Chamorro-Premuzic, T., & Frankiewicz, B. (2019). *Does hiring for cultural fit mean hiring for bias? Harvard Business Review*.
- ❖ Landers, R. N., & Marin, S. (2021). *Theory-driven game-based assessment of soft skills in organizations. Frontiers in Psychology*.
- ❖ IJCRT. (2023). *AI Resume Screening Technologies*.
- ❖ Greenfield, R., & Griffin, R. (2018). *AI and Social Skills Assessment*.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed solution is an **AI-based candidate recommendation system** designed to evaluate applicants holistically. It integrates advanced AI techniques to assess hard skills, soft skills, and social traits, providing an unbiased and comprehensive ranking of candidates for job roles. The system automates key recruitment tasks, including CV screening, technical skill evaluation, behavioral analysis, and social trait assessment, significantly improving efficiency and accuracy.

The system consists of four interconnected modules:

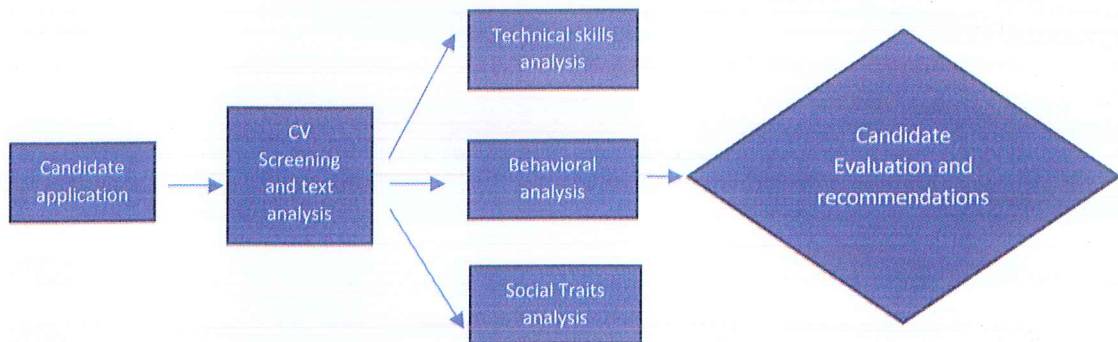
CV Screening and Text Analysis: Utilizes Natural Language Processing (NLP) to extract relevant data from CVs, rank candidates based on job-specific criteria, and identify potential mismatches.

Technical Skills Assessment: Deploys automation tools to simulate real-world tasks, evaluate performance, and provide objective scoring of job-related competencies.

Behavioral Analysis: Analyzes candidates' responses to hypothetical scenarios using sentiment analysis and decision-making models to assess traits like adaptability and emotional intelligence.

Social Traits Evaluation: Uses AI-driven tools such as facial recognition, voice analysis, or interactive tasks to evaluate interpersonal and teamwork skills.

These modules work together to provide a holistic view of each candidate, ensuring fair and efficient hiring.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

The development of an AI-based candidate recommendation system requires a multidisciplinary approach, combining expertise in human resources, artificial intelligence, and data science.

1. **Specialized Domain Expertise**

- **Human Resources (HR) and Recruitment:** Expertise in recruitment processes is essential to define the key evaluation criteria for technical and soft skills. Understanding job-specific requirements ensures that the AI models align with real-world hiring needs.
- **Artificial Intelligence (AI) and Machine Learning (ML):** Specialized knowledge in AI algorithms, including Natural Language Processing (NLP) for CV analysis, supervised and unsupervised learning for data classification, and computer vision for behavioral and social analysis, is crucial for system development.
- **Behavioral Psychology:** Insights into personality traits, decision-making, and emotional intelligence are necessary to design and interpret behavioral and social trait analyses effectively.

2. **Knowledge Requirements**

- **Natural Language Processing:** For processing unstructured text in CVs and extracting relevant features.
- **Automated Testing Systems:** Knowledge of designing task-based simulations for evaluating technical skills.
- **Sentiment Analysis and Emotion Recognition:** Understanding how AI can analyze textual and visual data for behavioral assessments.
- **Data Privacy and Ethics:** Ensuring compliance with regulations such as GDPR for handling sensitive candidate data.

3. **Data Requirements**

- **Candidate CV Dataset:** A large dataset of anonymized CVs, labeled with relevant features for training NLP models.
- **Behavioral Data:** Datasets including hypothetical scenario responses or annotated behavioral traits for supervised learning.
- **Social Trait Data:** Video and audio datasets annotated for communication, teamwork, and leadership traits.
- **Technical Skill Assessment Data:** Performance metrics from task-based simulations across various roles.

8. Objectives and Novelty

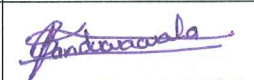
Main Objective The primary objective of this research is to develop an AI-based candidate recommendation system that evaluates candidates holistically by integrating assessments of technical skills, behavioral responses, social traits, and qualifications extracted from CVs. The system aims to automate and enhance the recruitment process, providing unbiased, efficient, and accurate recommendations to help organizations identify the most suitable candidates for job roles.			
Member Name	Sub Objective	Tasks	Novelty
S.U.D.A Gunasekara	CV Screening and Text Analysis To automate the extraction and evaluation of candidate qualifications, skills, and experience from CVs using Natural Language Processing (NLP). This component moves beyond traditional keyword-matching systems by implementing context-aware models. It combines Natural Language Processing (NLP) with semantic similarity techniques to evaluate CVs holistically. The novelty lies in the ability to dynamically	- Collect and preprocess a dataset of candidate CVs. - Implement NLP techniques to extract relevant features such as education, experience, and skills. - Develop a ranking algorithm to match candidates with job-specific requirements. - Optimize the model for accuracy and minimize bias in text analysis.	Introduces a context-aware CV analysis model that not only identifies relevant keywords but also evaluates the overall relevance of CVs based on job descriptions, reducing dependency on traditional keyword matching.

	rank candidates by understanding contextual relationships between skills and job descriptions, minimizing redundancy and bias in CV screening		
R.S.K De Silva	Technical Skills Assessment To design an automated system for evaluating candidates' technical skills through job-relevant task simulations. The proposed model integrates adaptive simulation techniques that adjust task complexity based on candidates' performance. This approach creates realistic workplace scenarios and uses real-time analytics to provide customized assessments, offering a more precise evaluation of technical proficiency. Unlike static testing methods, this model dynamically adapts, making it highly scalable for different roles and skill levels	• Design task-based simulations for specific job roles (e.g., coding challenges or troubleshooting exercises). • Develop automated grading systems using machine learning models to evaluate performance. • Test the system with datasets of candidate performance metrics. • Ensure fairness by standardizing evaluation criteria.	• Introduces adaptive simulations that adjust complexity based on candidate responses, providing a dynamic evaluation tailored to individual skill levels.

K.T.A.D.K Kasthurirathna	Behavioral Analysis To analyze candidates' responses to hypothetical scenarios to assess their behavioral traits, such as adaptability and decision-making. This component uses a multi-modal sentiment analysis framework that incorporates textual, tonal, and contextual data to assess candidates' behavioral traits. It captures nuanced emotional and cognitive responses to situational prompts, enabling a deeper evaluation of decision-making and adaptability. Unlike existing models, this framework integrates real-time sentiment and response analytics, providing actionable insights for recruiters while addressing bias through diverse training datasets	• Create a dataset of behavioral scenarios with annotated responses. • Develop sentiment analysis and decision-making models to evaluate candidate responses. • Implement a scoring mechanism for traits like emotional intelligence and problem-solving. • Test and refine the models for interpretability and accuracy.	• Utilizes multi-modal sentiment analysis, combining text, tone, and context to deliver deeper insights into behavioral responses.
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B.D.A.M Kumara	Social Traits Analysis To evaluate candidates' social traits, such as communication, teamwork, and leadership, using AI-driven methods. By combining AI-driven non-verbal cue analysis (e.g., facial recognition and voice modulation analysis) with interactive role-play simulations, this component offers a pioneering approach to evaluating interpersonal skills. The system interprets micro-expressions and vocal tone shifts to assess leadership, teamwork, and communication abilities in real-time. The novelty lies in the fusion of interaction analytics with machine learning models, which creates a comprehensive profile of candidates' social traits.	• Develop interactive tasks or role-playing simulations to evaluate interpersonal skills. • Use facial recognition and voice analysis for communication style assessment. • Train machine learning models on datasets annotated for social traits. • Integrate findings into a scoring mechanism for comprehensive evaluations.	• Combines real-time interaction analysis with AI to evaluate non-verbal cues, such as facial expressions and tone, for a deeper understanding of social dynamics.
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9. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Prof.	Samantha	Thelijjagoda	
Co-Supervisor	Ms.	Poozha	Panduawala	 09.12.2024
External Supervisor	Prof.	Nishantha	Giguruwa	/Sgned Nishantha Giguruwa
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	✓	No	
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- b) Does the proposed topic exhibit novelty?

Yes	✓	No	
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes	✓	No	
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	✓	No	
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

Recommended.

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	✓
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Research componets of all members must be clearly addressed.
 As intructed at the TAF evaluation, address the concern above and present again.

Staff Member's Name	Signature
Samantha Thelijjagoda Poorna Panduwawala	

*Important:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.